



# Modelling Financial Markets

## Seminar 12: Options and the Volatility Surface — Hands-on



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## Acronyms used in this material

- ▣ AI: Artificial Intelligence
- ▣ AR: AutoRegressive (model)
- ▣ ATM: At-The-Money
- ▣ CI: Confidence Interval
- ▣ CRR: Cox–Ross–Rubinstein (binomial tree)
- ▣ DM: Diebold–Mariano (test)
- ▣ DVOL: Deribit Volatility Index
- ▣ GMM: Generalized Method of Moments
- ▣ HAC: Heteroskedasticity and Autocorrelation Consistent (standard errors)
- ▣ HAR-RV: Heterogeneous AutoRegressive model of Realised Volatility
- ▣ MSE: Mean Squared Error
- ▣ MZ: Mincer–Zarnowitz (regression)
- ▣ OLS: Ordinary Least Squares
- ▣ OTM: Out-of-The-Money
- ▣ QLIKE: Quasi-LIKElihood (loss function)
- ▣ SVI: Stochastic Volatility Inspired (parameterisation)
- ▣ UTC: Coordinated Universal Time (Temps Universel Coordonné)
- ▣ VIX: CBOE Volatility IndeX
- ▣ VIX3M: Cboe 3-Month Volatility Index
- ▣ VRP: Variance Risk Premium
- ▣ XFG: Quantlet prefix of Applied Quantitative Finance (Härdle, Chen & Overbeck)

# Outline

Introduction

Part A: Derivations

Part B: Estimation and Inference

Part C: Open Question and AI Critique

Wrap-up

References

## Seminar Objectives

### By the end of this seminar you can

- ▣ Derive no-arbitrage restrictions, the Black–Scholes equation, the Greeks and the variance-swap strike, and check them on data
- ▣ Build delta and delta-gamma hedges and measure the hedging error with a confidence interval
- ▣ Fit a smile and a risk-neutral density with bootstrap bands, and test forecasts and the variance premium with robust inference

- ▣ Background: Lecture 12; textbook [Franke, Härdle & Hafner \(2019\)](#)

### Structure

- ▣ **Part A:** eight derivations, each checked on numbers (A1–A8)
- ▣ **Part B:** eight computational exercises with inference (B1–B8)
- ▣ **Part C:** an open question, a seed for the team project (C1), and an AI answer to critique (C2)

## How to Work in This Seminar

- ▣ **[Solved]** problems show the full worked solution; **[Proposed]** problems show only the task: their solutions are discussed in class
- ▣ Every proposed problem names its **model**: the solved problem that uses the same technique
  - ▶ Every Part B exercise ends with interpretation questions
- ▣ Volatility in annualised %; “vol points” = percentage points of volatility; time in years unless stated
- ▣ Overlapping windows are autocorrelated: always use Newey–West standard errors or a block bootstrap [Newey & West \(1987\)](#), [Künsch \(1989\)](#)

## Solved and Proposed Problems

### Solved: worked examples in this deck

- Part A: A1 (bounds and convexity in the strike), A3 (Black–Scholes equation and hedging P&L), A5 (one identity for the Greeks), A7 (implied volatility: existence and a starting value)
- Part B: B1 (hedging error with bootstrap intervals), B3 (SVI fit of a Bitcoin smile), B5 (variance risk premium of the S&P 500)

### Proposed: you solve them

- Part A: A2 (parity as a regression; model: A1), A4 (tree convergence; model: A3), A6 (delta-gamma hedge; model: A5), A8 (variance-swap strike and its errors; model: A1)
- Part B: B2 (delta-hedged S&P 500 options; model: B1, B5), B4 (a VIX for Bitcoin; model: A8, B3), B6 (is the VIX an efficient forecast?; model: B5), B7 (Bitcoin premium; model: B5), B8 (density with bands; model: B3)
- Part C: C1, open question and project seed; C2, critique of an AI answer (Black–Scholes and parity)

## Setup: Python Environment and Data

-  **Open in Colab:** `chapter12_seminar_notebook.ipynb`
- Google Colab (recommended) or local Python  $\geq 3.10$  with `numpy`, `pandas`, `scipy`, `statsmodels`, `matplotlib`
- Series: S&P 500 and VIX (1990–2026), VIX3M (2007–2026); daily closes up to 18 September 2026
- Bitcoin: daily closes; DVOL (Deribit, 2021–2026); the Bitcoin option chain of the lecture (Deribit snapshot)
- Two series together: prices joined on common days first, then returns

## Formulas Used Today

- Parity:  $C - P = Se^{-q\tau} - Ke^{-r\tau}$ ; binomial:  $q = (R - d)/(u - d)$ ,  $C = [qC_u + (1 - q)C_d]/R$  Cox, Ross & Rubinstein (1979)
- Black-Scholes Black & Scholes (1973):  $C = S\Phi(d_1) - Ke^{-r\tau}\Phi(d_2)$ ,  $d_{1,2} = [\ln(S/K) + (r \pm \sigma^2/2)\tau]/(\sigma\sqrt{\tau})$
- Greeks:  $\Delta = \Phi(d_1)$ ,  $\Gamma = \varphi(d_1)/(S\sigma\sqrt{\tau})$ , vega  $= S\varphi(d_1)\sqrt{\tau}$ ,  $\Theta = -S\varphi(d_1)\sigma/(2\sqrt{\tau}) - rKe^{-r\tau}\Phi(d_2)$
- Implied variance Cboe (2025):  $\sigma^2 = \frac{2}{T} \sum_i \frac{\Delta K_i}{K_i^2} e^{rT} Q(K_i) - \frac{1}{T} \left( \frac{F}{K_0} - 1 \right)^2$
- SVI Gatheral & Jacquier (2014):  $w(k) = a + b \left( \rho(k - m) + \sqrt{(k - m)^2 + s^2} \right)$ ,  $w = \sigma_{imp}^2 \tau$ ,  $k = \ln(K/F)$
- Variance risk premium Carr & Wu (2009):  $VRP_t = IV_t^2 - RV_{t,t+h}$ ,  $RV_{t,t+h} = \frac{A}{h} \sum_{i=1}^h r_{t+i}^2$  ( $A$  = periods per year)



## A1: Bounds and Convexity in the Strike [Solved]

### □ Context and data

- ▶ European calls on a non-dividend-paying asset, price  $C(K)$  for strike  $K$ , maturity  $\tau$ , continuous rate  $r$ ; no arbitrage, no frictions
- ▶ Data for (d): Bitcoin calls expiring 30 October 2026 (Deribit snapshot of the lecture, 22 September 2026), BTC prices converted to USD with the Bitcoin index  $S = 85\,265$  (the Deribit convention); forward  $F = 85\,742$ , so  $e^{r\tau} = F/S$

### □ Tasks

- ▶ (a) Prove  $\max(S - Ke^{-r\tau}, 0) \leq C \leq S$  by comparing the pay-offs of two portfolios in every state at  $T$
- ▶ (b) For  $K_1 < K_2 < K_3$  with  $K_2 = \lambda K_1 + (1 - \lambda)K_3$ , prove  $\lambda C(K_1) + (1 - \lambda)C(K_3) \geq C(K_2)$ , i.e.  $C$  is convex in  $K$
- ▶ (c) Show that the butterfly  $[C(K - h) - 2C(K) + C(K + h)]/h^2$  tends to  $e^{-r\tau}q(K)$ , with  $q$  the risk-neutral density [Breedon & Litzenberger \(1978\)](#)
- ▶ (d) Compute the butterfly 84/85/86 thousand USD from mark prices, the implied density, the triangularly weighted mass  $e^{r\tau}$  butterfly/ $h$  and  $P^Q(84\,000 < S_T < 86\,000)$ ; then count, over all 732 equally spaced call triples, how many violate convexity at mark prices and at executable prices (wings at the ask, body at the bid)

### □ Report and interpret

- ▶ The bounds and the convexity proof; the butterfly value,  $q(85\,000)$  and the violation counts; is any violation an arbitrage you could trade?

## A1: Solution (1/2) [Solved]

- (a) Lower bound: a call plus  $Ke^{-r\tau}$  in cash is worth  $\max(S_T, K) \geq S_T$  at  $T$ , so it cannot cost less than the share:  
 $C + Ke^{-r\tau} \geq S$ 
  - ▶ The call pays  $(S_T - K)^+ \geq 0$ , so  $C \geq 0$ ; together  $C \geq \max(S - Ke^{-r\tau}, 0)$
  - ▶ Upper bound: the share pays  $S_T \geq (S_T - K)^+$  in every state, so  $C \leq S$
- (b) For fixed  $S_T$  the map  $K \mapsto (S_T - K)^+$  is convex, so  $\lambda(S_T - K_1)^+ + (1 - \lambda)(S_T - K_3)^+ \geq (S_T - K_2)^+$  in every state
  - ▶ A portfolio that pays at least as much in every state costs at least as much today:  $\lambda C(K_1) + (1 - \lambda)C(K_3) \geq C(K_2)$
- (c)  $C(K) = e^{-r\tau} \int_K^\infty (x - K)q(x) dx$ , so  $\partial_K C = -e^{-r\tau} \int_K^\infty q(x) dx$  and  $\partial_K^2 C = e^{-r\tau} q(K)$ 
  - ▶ The butterfly quotient is the central second difference, so it converges to  $e^{-r\tau} q(K)$ : convexity is exactly  $q \geq 0$

## A1: Solution (2/2) [Solved]

- (d) Mark prices in USD:  $C(84\,000) = 4957.26$ ,  $C(85\,000) = 4414.38$ ,  $C(86\,000) = 3922.98$ 
  - ▶ Butterfly =  $4957.26 - 2 \cdot 4414.38 + 3922.98 = 51.47$  USD, positive as required
  - ▶  $e^{r\tau} = F/S = 1.0056$ :  $q(85\,000) \approx 1.0056 \cdot 51.47 / 1\,000^2 = 5.18 \cdot 10^{-5}$  per USD
  - ▶ The butterfly pays  $(h - |S_T - K|)^+$ :  $e^{r\tau}$  butterfly/ $h = 5.2\%$  is a triangularly weighted mass  $\approx h q(K)$ ;  $P^Q(84\,000 < S_T < 86\,000) \approx 2h q(K) = 10.4\%$
- All triples: 7 of 732 mark butterflies are negative, all among deep in-the-money calls, the worst  $-11.80$  USD
  - ▶ The median bid–ask spread at those strikes is 255.80 USD; at executable prices the number of negative butterflies is 0
- Interpretation: no tradable arbitrage; but a density estimated from raw marks would be negative in places, which is why the lecture smooths the smile (SVI) before differentiating

## A2: Parity as a Regression: the Implied Forward [Proposed]

Solution discussed in class

### □ Context and data

- ▶ European calls and puts with the same expiry and strikes  $K_1, \dots, K_n$ ; discount factor  $D = e^{-r\tau}$  and forward  $F$  unknown
- ▶ Data: the expiry of A1 (30 October 2026), USD mid prices  $(bid + ask)/2 \times S$  ( $S$  the Bitcoin index, the Deribit convention), only strikes where both the call and the put have a bid and an ask

### □ Tasks

- ▶ (a) Show from put–call parity that  $C(K) - P(K) = DF - DK$  for every strike
- ▶ (b) Write the OLS regression  $C_i - P_i = a + bK_i + e_i$ ; derive  $\hat{D}$ ,  $\hat{F}$  from  $(\hat{a}, \hat{b})$  and the delta-method standard error of  $\hat{F}$  with heteroskedasticity-robust covariance
- ▶ (c) Estimate on the data; compare  $\hat{F}$  with the exchange forward in standard errors

### □ Report and interpret

- ▶  $\hat{D}$ ,  $\hat{F}$  with standard errors, the number of strikes; which USD interest rate does  $\hat{D}$  imply, and why might  $\hat{F}$  differ from the exchange forward?
- ▶ Model: A1 (portfolios with identical pay-offs)

## A3: The Black–Scholes Equation and the Hedging P&L [Solved]

### □ Context and data

- ▶ Price  $dS_t = \mu S_t dt + \sigma S_t dW_t$ , constant  $r$ , continuous trading without costs; option value  $V(S, t)$ , twice differentiable [Black & Scholes \(1973\)](#)
- ▶ Numbers for (c): the call of A5 ( $S = K = 100$ , six months,  $r = 3\%$ ), sold at  $\sigma_{imp} = 25\%$ , gamma  $\Gamma = 0.0222$ ; 252 trading days a year

### □ Tasks

- ▶ (a) Replicate  $V$  with a self-financing portfolio of  $\Delta$  shares and cash  $B$  at  $r$  ( $d\Pi = \Delta dS + rB dt$ ); apply Itô's lemma to  $V$ , choose  $\Delta$  to remove the  $dW_t$  term and use no arbitrage to derive the Black–Scholes equation
- ▶ (b) A dealer sells the option at  $\sigma_{imp}$ , holds  $\Delta = \partial_S V^{imp}$  shares and cash at  $r$ , while  $S$  moves with  $\sigma_{real}$ ; derive the instantaneous P&L
- ▶ (c) Compute today's P&L rate per trading day if  $\sigma_{real} = 30\%$ , then annualise it with  $S$  and  $\Gamma$  frozen at today's values

### □ Report and interpret

- ▶ The equation, the P&L formula and the two numbers; who gains and why does the P&L depend on the path of  $S$ ?

## A3: Solution (1/2) [Solved]

- (a) Itô:  $dV = (V_t + \mu SV_S + \frac{1}{2}\sigma^2 S^2 V_{SS}) dt + \sigma SV_S dW_t$ ; the self-financing  $\Pi = \Delta S + B$ , started at  $\Pi = V$ , has  $d\Pi = \Delta dS + rB dt$
- $d(V - \Pi) = (V_t + \frac{1}{2}\sigma^2 S^2 V_{SS} - rB) dt + (V_S - \Delta)(\mu S dt + \sigma S dW_t)$ ; with  $\Delta = V_S$  both the risk and  $\mu$  disappear
- A riskless position that costs 0 earns 0:  $V_t + \frac{1}{2}\sigma^2 S^2 V_{SS} = rB = r(V - SV_S)$ , i.e.  $V_t + \frac{1}{2}\sigma^2 S^2 V_{SS} + rSV_S - rV = 0$
- (b) The self-financing book  $-V + \Delta S + (V - \Delta S)$  in cash changes by  $-dV + \Delta dS + r(V - \Delta S) dt$ , plus interest on the P&L already accumulated
- Itô with the true volatility:  $dV = V_t dt + \Delta dS + \frac{1}{2}\Gamma\sigma_{real}^2 S^2 dt$ ; the equation at  $\sigma_{imp}$  gives  $V_t = rV - rS\Delta - \frac{1}{2}\sigma_{imp}^2 S^2 \Gamma$

## A3: Solution (2/2) [Solved]

- Substituting, the  $dS$  and  $r$  terms cancel:  $d \text{P\&L} = r \text{P\&L} dt + \frac{1}{2} \Gamma S^2 (\sigma_{imp}^2 - \sigma_{real}^2) dt$ ; the second term is the excess gain
- (c)  $\frac{1}{2} \cdot 0.02223 \cdot 100^2 \cdot (0.25^2 - 0.30^2) \cdot \frac{1}{252} = \frac{1}{2} \cdot 0.02223 \cdot 10^4 \cdot (-0.0275)/252 = -0.0121$  per option and trading day
- Annualised with  $S$  and  $\Gamma$  frozen:  $-3.06$ , about 39% of the call price 7.76; a rate, not the realised P&L of this six-month option, since  $\Gamma S^2$  changes along the path
- Interpretation: the seller loses when realised volatility exceeds implied; the loss accrues through  $\Gamma$ , largest near the strike and near expiry, so two paths with the same realised variance give different P&L

## A4: The Binomial Tree Converges to Black–Scholes [Proposed]

Solution discussed in class

### □ Context and data

- ▶ Cox–Ross–Rubinstein (CRR) tree with  $N$  steps of length  $\Delta t = T/N$  Cox, Ross & Rubinstein (1979); European call of the lecture:  $S = K = 100$ ,  $T = 1$ ,  $r = 5\%$ ,  $\sigma = 20\%$

### □ Tasks

- ▶ (a) Show that  $u = e^{\sigma\sqrt{\Delta t}}$ ,  $d = 1/u$ ,  $q = (e^{r\Delta t} - d)/(u - d)$  match the risk-neutral mean exactly and the variance of  $\ln(S_{t+\Delta t}/S_t)$  up to terms of order  $\Delta t^2$
- ▶ (b) Price the call with  $N = 50, 51, 100, 101, 200, 201$  steps and compute  $N(C_N - C_{BS})$  for each
- ▶ (c) Average the prices for  $N = 200$  and  $N = 201$  and compare the error with that of  $N = 200$

### □ Report and interpret

- ▶ A table of  $C_N$  and  $N(C_N - C_{BS})$ ; the order of the error, the reason for the even/odd oscillation, and why averaging helps
- ▶ Model: A3 (from the tree to the continuous-time equation)



## A5: One Identity for All the Greeks [Solved]

### □ Context and data

- ▶ Black–Scholes call  $C = S\Phi(d_1) - Ke^{-r\tau}\Phi(d_2)$ ,  $d_{1,2} = [\ln(S/K) + (r \pm \sigma^2/2)\tau]/(\sigma\sqrt{\tau})$ ;  $\varphi$  = standard Normal density
- ▶ Numbers for (c):  $S = K = 100$ , six months ( $\tau = 0.5$ ),  $r = 3\%$ ,  $\sigma = 25\%$ , no dividends

### □ Tasks

- ▶ (a) Prove  $S\varphi(d_1) = Ke^{-r\tau}\varphi(d_2)$
- ▶ (b) Use (a) to derive  $\Delta = \partial_S C = \Phi(d_1)$  and vega  $= \partial_\sigma C = S\varphi(d_1)\sqrt{\tau}$ ; then show vega  $= \Gamma S^2\sigma\tau$  with  $\Gamma = \varphi(d_1)/(S\sigma\sqrt{\tau})$
- ▶ (c) Compute  $d_1$ ,  $d_2$ , both sides of (a),  $\Delta$ ,  $\Gamma$ , vega per unit of  $\sigma$  in two ways, the call and the put (from parity)

### □ Report and interpret

- ▶ The proofs and the numerical checks; what does vega  $= \Gamma S^2\sigma\tau$  imply for hedging gamma with options of the same expiry?

## A5: Solution (1/2) [Solved]

- (a)  $\varphi(d_2)/\varphi(d_1) = e^{(d_1^2 - d_2^2)/2}$  and  $d_1^2 - d_2^2 = (d_1 - d_2)(d_1 + d_2) = \sigma\sqrt{\tau} \cdot \frac{2 \ln(S/K) + 2r\tau}{\sigma\sqrt{\tau}}$
- Hence  $\varphi(d_2)/\varphi(d_1) = Se^{r\tau}/K$ , i.e.  $S\varphi(d_1) = Ke^{-r\tau}\varphi(d_2)$
- (b)  $\partial_S C = \Phi(d_1) + [S\varphi(d_1) - Ke^{-r\tau}\varphi(d_2)] \partial_S d_1 = \Phi(d_1)$ , since  $\partial_S d_1 = \partial_S d_2$  and the bracket is 0 by (a)
- Vega:  $S\varphi(d_1)\partial_\sigma d_1 - Ke^{-r\tau}\varphi(d_2)\partial_\sigma d_2 = S\varphi(d_1)\partial_\sigma(d_1 - d_2) = S\varphi(d_1)\sqrt{\tau}$
- And  $\Gamma S^2 \sigma \tau = \frac{\varphi(d_1)}{S\sigma\sqrt{\tau}} S^2 \sigma \tau = S\varphi(d_1)\sqrt{\tau} = \text{vega}$

## A5: Solution (2/2) [Solved]

- (c)  $\sigma\sqrt{\tau} = 0.1768$ ;  $d_1 = (0 + (0.03 + 0.03125) \cdot 0.5)/0.1768 = 0.1732$ ,  $d_2 = -0.0035$
- Identity:  $S\varphi(d_1) = 100 \cdot 0.3930 = 39.30$  and  $Ke^{-r\tau}\varphi(d_2) = 98.51 \cdot 0.3989 = 39.30$ : equal
- $\Delta = \Phi(0.1732) = 0.5688$ ;  $\Gamma = 0.3930/(100 \cdot 0.1768) = 0.0222$
- $\text{Vega} = 100 \cdot 0.3930 \cdot \sqrt{0.5} = 27.79$  and  $\Gamma S^2 \sigma \tau = 0.02223 \cdot 10^4 \cdot 0.25 \cdot 0.5 = 27.79$  per unit of  $\sigma$  (0.2779 per vol point)
- $C = 100 \cdot 0.5688 - 98.51 \cdot 0.4986 = 7.76$ ; parity:  $P = C - 100 + 98.51 = 6.27$
- Interpretation: for one expiry, vega and gamma are proportional, so a gamma-neutral book of same-expiry options is also vega-neutral, but only while the smile is flat (A6)

## A6: A Delta-Gamma Hedge [Proposed]

Solution discussed in class

### □ Context and data

- ▶ A dealer has **sold** 1 000 calls of A5 ( $S = K = 100$ , six months,  $r = 3\%$ ,  $\sigma = 25\%$ )
- ▶ Hedging instruments: a call with  $K = 110$ , same expiry, Black–Scholes price 3.90, and the stock; Greeks from the Black–Scholes formulas

### □ Tasks

- ▶ (a) Compute the delta and the gamma of the dealer's position
- ▶ (b) Find the number  $n_2$  of  $K = 110$  calls that makes the gamma zero, then the number of shares that makes the delta zero
- ▶ (c) Compute the vega of the hedged book and explain the result

### □ Report and interpret

- ▶ The positions in the two instruments and the residual vega; is the book protected against a change in volatility?
- ▶ Model: A5

## A7: Implied Volatility: Existence, Uniqueness, Starting Value [Solved]

### □ Context and data

- ▶ European call, Black–Scholes price  $BS(\sigma)$ ; market price 4.50 for  $S = K = 100$ , three months ( $\tau = 0.25$ ),  $r = 2\%$

### □ Tasks

- ▶ (a) Prove that  $\sigma \mapsto BS(\sigma)$  is continuous and strictly increasing, with limits  $(S - Ke^{-r\tau})^+$  as  $\sigma \rightarrow 0$  and  $S$  as  $\sigma \rightarrow \infty$ ; conclude when an implied volatility exists
- ▶ (b) Derive the at-the-money-forward approximation  $C \approx S\sigma\sqrt{\tau}/\sqrt{2\pi}$  [Brenner & Subrahmanyam \(1988\)](#)
- ▶ (c) Use (b), with and without subtracting half the intrinsic value  $(S - Ke^{-r\tau})/2$ , as a starting value; then run one Newton step  $\sigma_1 = \sigma_0 - (BS(\sigma_0) - 4.50)/\mathcal{V}(\sigma_0)$  from  $\sigma_0 = 20\%$

### □ Report and interpret

- ▶ The two starting values,  $\sigma_1$  and the exact implied volatility; why does Newton converge so fast here, and when would it fail?

## A7: Solution (1/2) [Solved]

- (a)  $\partial_{\sigma} BS = S\varphi(d_1)\sqrt{\tau} > 0$  (A5): strictly increasing and continuous
- $\sigma \rightarrow 0$ :  $d_{1,2} \rightarrow \pm\infty$  according to the sign of  $\ln(S/K) + r\tau$ , so  $BS \rightarrow (S - Ke^{-r\tau})^+$ ;  $\sigma \rightarrow \infty$ :  $d_1 \rightarrow \infty$ ,  $d_2 \rightarrow -\infty$ , so  $BS \rightarrow S$
- By the intermediate value theorem a unique  $\sigma_{imp}$  exists iff the price lies strictly inside the no-arbitrage bounds of A1
- (b) If  $S = Ke^{-r\tau}$ :  $d_{1,2} = \pm\sigma\sqrt{\tau}/2$  and  $C = S[\Phi(\sigma\sqrt{\tau}/2) - \Phi(-\sigma\sqrt{\tau}/2)] \approx S\sigma\sqrt{\tau}\varphi(0) = S\sigma\sqrt{\tau}/\sqrt{2\pi}$

## A7: Solution (2/2) [Solved]

- (c) Without correction:  $\sigma \approx 4.50\sqrt{2\pi}/(100 \cdot 0.5) = 22.56\%$
- With correction:  $Ke^{-r\tau} = 99.501$ , half the intrinsic value 0.249,  $\sigma \approx (4.50 - 0.249)\sqrt{2\pi}/50 = 21.31\%$
- Newton:  $BS(20\%) = 4.2322$ ,  $\mathcal{V} = 19.848$ , so  $\sigma_1 = 0.20 - (4.2322 - 4.50)/19.848 = 21.3495\%$ ; exact 21.3495%
- Interpretation: near the money the price is almost linear in  $\sigma$ , so one Newton step gives four decimals, and the corrected approximation is already within 0.05 vol points
- Deep out of the money vega is close to 0 and Newton can overshoot; there, use a bracketing solver (Brent) inside the bounds

## A8: The Variance-Swap Strike and Its Estimation Error [Proposed]

Solution discussed in class

### □ Context and data

- ▶ Spanning formula of the lecture:  $f(S_T) = f(F) + f'(F)(S_T - F) + \int_0^F f''(K)(K - S_T)^+ dK + \int_F^\infty f''(K)(S_T - K)^+ dK$   
Bakshi, Kapadia & Madan (2003)
- ▶ Log-normal benchmark:  $F = 100$ ,  $r = 0$ , 90 days ( $T = 90/365$ ), flat volatility 20%, so the exact variance is 0.04; OTM prices from Black–Scholes

### □ Tasks

- ▶ (a) Apply the formula with  $f(S) = -2 \ln(S/F)$  and show  $K_{var} = \frac{2e^{rT}}{T} \left[ \int_0^F \frac{P(K)}{K^2} dK + \int_F^\infty \frac{C(K)}{K^2} dK \right]$  for continuous paths
- ▶ (b) Compute the Cboe sum with  $\Delta K = 5$  on  $[80, 120]$  Cboe (2025), a dense integral on  $[80, 120]$  and a dense integral on  $(0, \infty)$
- ▶ (c) Split the total error into discretisation (Cboe minus dense on  $[80, 120]$ ) and truncation (dense on  $[80, 120]$  minus exact)

### □ Report and interpret

- ▶ The three variances and volatilities and the two errors; which dominates here, and what changes for a longer maturity?
- ▶ Model: A1 (portfolios of options as prices of pay-off shapes)



## B1: Hedging Error and Rebalancing Frequency [Solved]

### ▣ Context and data

- ▶ Simulated data: 10 000 paths of geometric Brownian motion,  $S_0 = 100$ ,  $\mu = 8\%$ ,  $\sigma = 20\%$ , 252 steps over three months (seed 42)
- ▶ A three-month ATM call ( $K = 100$ ,  $r = 4\%$ ) is sold at its Black–Scholes price and delta-hedged at  $\sigma = 20\%$

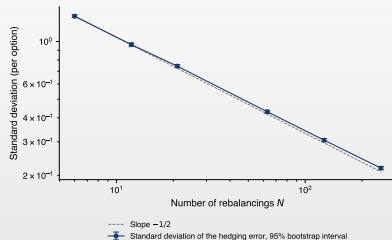
### ▣ Tasks

- ▶ (a) Rebalance  $N = 6, 12, 21, 63, 126, 252$  times; compute the standard deviation of the hedging error at expiry for each  $N$
- ▶ (b) Build 95% percentile bootstrap intervals for each standard deviation from 1 000 resamples of the paths [Efron \(1979\)](#)
- ▶ (c) Regress  $\ln(\text{s.d.})$  on  $\ln N$ ; give the slope with a bootstrap interval (the same 1 000 resamples)

### ▣ Report and interpret

- ▶ A table of s.d. and intervals and a log-log chart; is the slope consistent with  $-1/2$ ? How often must a dealer rebalance to keep the error below 10% of the premium? What would transaction costs change?

## B1: Solution (1/2) [Solved]



| $N$    | 6              | 12             | 21             | 63             | 126            | 252            |
|--------|----------------|----------------|----------------|----------------|----------------|----------------|
| s.d.   | 1.356          | 0.963          | 0.744          | 0.430          | 0.306          | 0.219          |
| 95% CI | [1.331, 1.381] | [0.945, 0.983] | [0.731, 0.758] | [0.422, 0.438] | [0.300, 0.312] | [0.215, 0.223] |

## B1: Solution (2/2) [Solved]

- Step 1: for each path, sell the call at the premium 4.49, buy  $\Delta_0$  shares, finance with cash at  $r$ ; at each rebalancing date reset  $\Delta$ ; at expiry, error = shares + cash - pay-off
- Step 2: the s.d. falls from 1.356 ( $N = 6$ ) to 0.219 ( $N = 252$ ); the intervals are narrow because 10 000 paths give a precise s.d.
- Step 3: slope  $-0.489$ , 95% CI  $[-0.496, -0.482]$ : close to  $-1/2$ , slightly above it at these finite  $N$  (the  $1/\sqrt{N}$  law is asymptotic)
- Reading: daily hedging leaves 9.6% of the premium, about the 10% target; four times a day halves it again (ratio 1.97)
- With a fixed fee per trade, costs grow like  $N$  while the error falls like  $1/\sqrt{N}$ ; the optimal frequency depends on the cost model and on the chosen trade-off between risk and cost

## B2: Delta-Hedged S&P 500 Options, 1990–2026 [Proposed]

Solution discussed in class

### □ Context and data

- ▶ Daily S&P 500 closes and VIX (GSPC.INDX, VIX.INDX in data/market), January 1990 to September 2026; interest and dividends ignored
- ▶ Every 21 trading days, sell a one-month ATM call priced with the VIX and hedge it daily at the same volatility (the stylised back-test of the lecture: VIX, not ATM quotes) [Bakshi & Kapadia \(2003\)](#); P&L in % of the index

### □ Tasks

- ▶ (a) Compute the mean monthly P&L and its Newey–West standard error with 3 lags; test  $H_0$ : mean = 0 at the 5% level
- ▶ (b) Build a 95% moving-block bootstrap interval for the mean (blocks of 6 months, 2 000 resamples) [Künsch \(1989\)](#)
- ▶ (c) Compute the share of positive months, the skewness and the share of the total gain lost in the five worst months

### □ Report and interpret

- ▶ The mean with both intervals and the tail statistics; is the premium significant? Why can a strategy with a high  $t$ -statistic still be dangerous? What would interest and dividends change?
- ▶ Model: B1 (bootstrap), B5 (Newey–West)

## B3: An SVI Smile for Bitcoin [Solved]

### □ Context and data

- ▶ Deribit snapshot of the lecture (`ch12_deribit_snapshot.csv`); expiry closest to 30 days: 2026-10-30, 38 days, 56 quoted OTM options (puts below the forward, calls above)
- ▶ Total variance  $w(k) = \sigma_{\text{mark}}^2 \tau$  at log-moneyness  $k = \ln(K/F)$ ; SVI  $w(k) = a + b(\rho(k - m) + \sqrt{(k - m)^2 + s^2})$  Gatheral & Jacquier (2014)

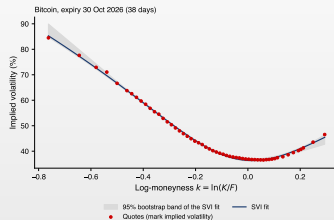
### □ Tasks

- ▶ (a) Fit the five SVI parameters by least squares on  $w$ , with  $b \geq 0$ ,  $|\rho| < 1$ ,  $s > 0$ ; report the fit error in vol points
- ▶ (b) Pairs bootstrap (300 resamples of the quotes, refit each time): 95% intervals for the ATM volatility and the skew  $IV(0.15) - IV(-0.15)$
- ▶ (c) Evaluate the butterfly function  $g(k)$  of Gatheral & Jacquier (2014) on  $k \in [-1.5, 1.5]$  and report its minimum

### □ Report and interpret

- ▶ Parameters, intervals and  $\min g$ ; which parameter controls the skew and what is its sign? Is the skew significantly different from 0? Why fit  $w$  and not the prices?

## B3: Solution (1/2) [Solved]



- (a)  $\hat{a} = -0.020$ ,  $\hat{b} = 0.102$ ,  $\hat{\rho} = -0.262$ ,  $\hat{m} = -0.079$ ,  $\hat{s} = 0.348$ ; root mean squared error 0.42 vol points

## B3: Solution (2/2) [Solved]

- (b) ATM volatility 36.37%, 95% CI [36.10, 36.87]; skew  $-2.11$ , 95% CI  $[-2.73, -1.52]$
- $\rho$  controls the tilt:  $\hat{\rho} < 0$  makes the put wing dearer; the skew interval excludes 0, so the tilt is significant at 5%
- (c)  $\min_k g(k) = 0.136 > 0$ : no butterfly arbitrage in this slice, so the implied density is positive
- Why  $w$ : it is comparable across strikes and nearly linear in the wings, while option prices span four orders of magnitude and would weight the fit towards the dearest options
- Caveat: the bootstrap resamples smoothed mark volatilities, so the intervals measure fitting noise, not bid–ask uncertainty

## B4: A VIX for Bitcoin [Proposed]

Solution discussed in class

### □ Context and data

- ▶ Same snapshot; the two expiries around 30 days (2026-10-09, 2026-10-30); USD mid price  $Q = \text{BTC mid} \times \text{index } S$  (the Deribit convention);  $F$  and  $e^{RT} = F/S$  from the Deribit forward

### □ Tasks

- ▶ (a) Compute the implied variance of each expiry with the Cboe formula [Cboe \(2025\)](#):  $K_0$  = first strike at or below  $F$ ; puts below  $K_0$ , calls above, the put–call average at  $K_0$ ; no zero bids, each wing stops after two consecutive zero bids;  $\Delta K_i$  = half the distance between neighbouring strikes, one-sided at the ends
- ▶ (b) Interpolate the total variances linearly in time to 30 days and take  $100\sigma_{30}$
- ▶ (c) Compare with the ATM volatilities of the two expiries and with DVOL at the same moment [Alexander & Imeraj \(2020\)](#)

### □ Report and interpret

- ▶ The two variances, the weight of the near expiry and the index; why is the index above both ATM volatilities, and which choices (strike cut-off, mid vs mark prices) could explain the gap to DVOL?
- ▶ Model: A8 (strip formula and its errors), B3 (the same snapshot)



## B5: The Variance Risk Premium of the S&P 500 [Solved]

### □ Context and data

- ▶ Daily S&P 500 and VIX closes, joined on common days, January 1990 to August 2026 (last window ending in September 2026)
- ▶ Ex-post premium  $VRP_t = VIX_t^2 - RV_{t,t+21}$ ,  $RV_{t,t+21} = \frac{252}{21} \sum_{i=1}^{21} r_{t+i}^2$  in %<sup>2</sup> Carr & Wu (2009); vol points:  $VIX_t - \sqrt{RV_{t,t+21}}$

### □ Tasks

- ▶ (a) Compute the mean VRP, its Newey–West standard error with 21 lags Newey & West (1987), a 95% interval and the naive standard error  $s/\sqrt{n}$
- ▶ (b) Repeat in vol points and compute the share of days with a positive premium
- ▶ (c) Repeat for 1990–2007 and 2008–2026, test each subperiod mean at the 5% level, then test their equality: regress  $VRP_t$  on a constant and a 2008–2026 dummy (Newey–West, 21 lags)

### □ Report and interpret

- ▶ A table by period; why is the naive standard error wrong here, and by how much? Is the premium stable across subperiods? Why is the variance premium less precise than the vol-point premium?

## B5: Solution (1/2) [Solved]

|           | $n$   | Mean VRP (% <sup>2</sup> ) | 95% CI         | Vol points (s.e.) | Positive |
|-----------|-------|----------------------------|----------------|-------------------|----------|
| 1990–2026 | 9 203 | 113.1                      | [72.7, 153.6]  | 4.09 (0.27)       | 86%      |
| 1990–2007 | 4 516 | 149.3                      | [125.2, 173.5] | 4.60 (0.25)       | 88%      |
| 2008–2026 | 4 687 | 78.3                       | [2.7, 153.9]   | 3.60 (0.46)       | 84%      |

- Step 1: build  $RV_{t,t+21}$  from the next 21 log returns, then  $VRP_t$  on every day with a complete window
- Step 2: regress  $VRP_t$  on a constant with HAC (Bartlett, 21 lags: the overlap; check longer bandwidths) standard errors; interval = mean  $\pm 1.96$  s.e.

## B5: Solution (2/2) [Solved]

- (a) Full sample:  $t = 113.1/20.6 = 5.49$ : reject  $H_0$  at 5%; naive s.e. 5.3 vs Newey–West 20.6: 3.9 times too small, because consecutive windows share 20 of 21 days
- (b) In vol points the premium is 4.09 (s.e. 0.27), positive on 86% of days
- (c) 1990–2007:  $t = 12.12$ ; 2008–2026:  $t = 2.03$ : both reject at 5%, but after 2008 barely, because of the crisis windows of 2008 and 2020
- Equality: difference  $-71.0\% ^2$  (s.e. 40.5,  $t = -1.76$ ,  $p = 0.08$ ); in vol points  $-1.01$  ( $t = -1.92$ ,  $p = 0.06$ )
- Reading: squaring magnifies the crisis windows, so the variance premium has heavier tails and a wider interval than the vol-point premium; the premium is positive in both subperiods, but the difference in size is not significant at 5%: the data cannot tell whether it changed

## B6: Is the VIX an Efficient Forecast? [Proposed]

Solution discussed in class

### □ Context and data

- ▶ The daily series of B5:  $RV_{t,t+21}$  and  $VIX_t^2$ , 1990–2026; for (d) also daily squared returns to build HAR-RV regressors (1, 5 and 22-day averages) [Corsi \(2009\)](#)

### □ Tasks

- ▶ (a) Estimate  $RV_{t,t+21} = \alpha + \beta VIX_t^2 + \varepsilon_t$  [Mincer & Zarnowitz \(1969\)](#) with Newey–West errors (21 lags); test  $\alpha = 0$  and  $\beta = 1$  separately and jointly (Wald, 5%); report  $\text{corr}(\hat{\alpha}, \hat{\beta})$
- ▶ (b) Add  $RV_{t-21,t}$ ; then re-estimate the regression in logs,  $\ln RV$  on  $\ln VIX^2$ , and test  $\beta = 1$
- ▶ (c) Instrument  $\ln VIX_t^2$  with  $\ln VIX_{t-21}^2$  [Christensen & Prabhala \(1998\)](#) (just-identified GMM, HAC covariance); report the first-stage  $t$  and test  $\beta = 1$
- ▶ (d) From 2000, re-estimate every 21 days on targets already observed; compare  $VIX^2$  with HAR-RV by QLIKE loss and a Diebold–Mariano test with HAC (21 lags) [Diebold & Mariano \(1995\)](#), [Patton \(2011\)](#)

### □ Report and interpret

- ▶ All estimates and tests with decisions at 5%; why can both  $t$ -statistics be small while the joint test rejects? How do the conclusions change in logs and with instruments, and what must the instrument satisfy? Does in-sample significance survive out of sample?
- ▶ Model: B5 (Newey–West inference)

## B7: The Variance Risk Premium of Bitcoin [Proposed]

Solution discussed in class

### □ Context and data

- ▶ DVOL (Deribit), March 2021 to August 2026, and daily Bitcoin closes (BTC-USD.CC); realised volatility over the next 30 calendar days,  $A = 365$

### □ Tasks

- ▶ (a) Compute the mean premium  $DVOL_t - \sqrt{RV_{t,t+30}}$  in vol points, its Newey–West standard error with 30 lags, and test it at 5%
- ▶ (b) Compute the share of positive days and the ratio mean implied / mean realised volatility; compare both with the S&P 500 of B5
- ▶ (c) Count the non-overlapping 30-day periods in the sample

### □ Report and interpret

- ▶ Premium, test, ratios and effective sample size; is the Bitcoin premium larger or smaller than the equity premium, and in which unit? How much can five years of data tell us?
- ▶ Model: B5

## B8: A Risk-Neutral Density with Confidence Bands [Proposed]

Solution discussed in class

### □ Context and data

- ▶ Deribit snapshot of the lecture; expiry closest to 90 days: 2026-12-25, 94 days, 58 OTM quotes; SVI fitted as in B3; forward (undiscounted) prices

### □ Tasks

- ▶ (a) Compute the density  $q(K) = \partial^2 C / \partial K^2$  from the SVI call prices (Black-76 with  $F$ ) on a fine strike grid [Breedon & Litzenberger \(1978\)](#)
- ▶ (b) Compute  $P^Q(S_T < 0.8F)$  and  $P^Q(S_T > 1.2F)$  with 95% pairs-bootstrap intervals (300 refits) and the log-normal values at the ATM volatility
- ▶ (c) Report where the quotes end in  $K/F$  and which part of each probability relies on extrapolated wings

### □ Report and interpret

- ▶ The two probabilities with intervals vs log-normal; is the crash probability significantly above or below the log-normal one? What does the bootstrap band not measure?
- ▶ Model: B3 (SVI fit and pairs bootstrap)

## C1: Is the Variance Risk Premium a Tradable Signal?

### Question

- Would you sell variance on the S&P 500 or on Bitcoin every month, and when would you stop? Carr & Wu (2009), Bollerslev, Tauchen & Zhou (2009)
- Instrument: a synthetic variance swap with vega notional 1,  $P\&L = (IV^2 - RV)/(2 IV)$  vol points per month
- S&P 500 with the VIX (1990–2026); Bitcoin with DVOL (2021–2026)

### What a good answer contains

- Sharpe ratio with a block-bootstrap interval, skewness, worst month, maximum drawdown
- A timing rule, e.g. skip months that start in backwardation (VIX above VIX3M); is it better out of sample?
- Comparison with the delta-hedged option of the lecture; position sizing and margin
- The ex-ante VRP ( $VIX_t^2/12 - RV_t$ ) as a predictor of 1–12-month excess returns, with Hodrick (1992) errors and a bootstrap under the null Bollerslev, Tauchen & Zhou (2009), Hodrick (1992)

## C2: Critique an AI Answer [Proposed]

- A student asks an AI assistant: “Price a European call with 90 calendar days to expiry on a non-dividend-paying stock with  $S = 100$ ,  $K = 105$ , volatility 20%, continuous rate  $r = 4\%$ ; then give the put from put–call parity”
- The assistant answers with this code:

```
import numpy as np
from scipy.stats import norm
S, K, r, sigma = 100.0, 105.0, 0.04, 20          # sigma = 20%
T = 90 / 252                                     # 90 days
d1 = (np.log(S / K) + (r + 0.5 * sigma**2) * T) / (sigma * np.sqrt(T))
d2 = d1 - sigma * np.sqrt(T)
call = S * norm.cdf(d1) - K * np.exp(-r * T) * norm.cdf(d2)
put = call + S - K * np.exp(-r * T)             # put-call parity
print(f"call_{S}={call:.2f},_{K}put_{S}={put:.2f}")
```

- “The code follows Black–Scholes exactly; the put comes from parity, so no second formula is needed”



## C2: Critique an AI Answer, Task [Proposed]

Solution discussed in class

### Task

- The answer contains **three** errors; everything else is correct
- For each error
  - ▶ say what is wrong and why
  - ▶ say how you would detect it: a no-arbitrage bound, a unit check, the lecture, a second method
  - ▶ give the corrected line
- Run the corrected code; report  $C$  and  $P$  and check put-call parity numerically
- Log the three errors as in `AI_ERRORS.md`: what was asked, what the AI answered, what was wrong, how you found out, the correct version

### Interpretation

- Before running the code: what should a 90-day call 5% out of the money cost, roughly: 0.2, 2 or 20?
- Which error would still give a plausible-looking price? Which check catches it?
- Why is “the put comes from parity” not a check of the call price?

## Common Pitfalls

- ▣ Forgetting dividends or the interest rate in parity: a false arbitrage
- ▣ Treating the risk-neutral probability as a forecast
- ▣ Quoting vega or theta without the unit (per vol point? per day? per year?)
- ▣ Newton's method on deep OTM options ( $\text{vega} \approx 0$ ): use a bracketing solver
- ▣ Naive standard errors on overlapping windows:  $t$ -statistics several times too large
- ▣ Reading a high Sharpe ratio of a short-volatility strategy without its skewness and worst month

## Take-Home Assignment (Team Project)

- ▣ Take a new Deribit snapshot for Ether (ETH) options and build the smile of three expiries with SVI, with bootstrap intervals
- ▣ Compute a 30-day VIX-type index and the risk-neutral density; compare with Bitcoin [Pele et al. \(2023\)](#)
- ▣ Delta-hedge a sold one-week ATM Bitcoin option on historical paths since March 2021 [Matic, Packham & Härdle \(2023\)](#)
  - ▶ Price and hedge each option at the DVOL of its trade date (the information then available),  $r = 0$ , daily rebalancing at the 00:00 UTC closes, non-overlapping weekly windows
  - ▶ Report the mean and standard deviation of the error in % of the premium, with moving-block bootstrap intervals; justify every choice you change
- ▣ Deliver a Quantlet folder (code, Metainfo, charts) and three slides: numbers, interpretation, limitations
- ▣ Log every AI error you catch in `AI_ERRORS.md`, as in C2: what was asked, what was wrong, how you found out, the correct version

## Further Study on Quantinar

### Advanced courses on quantinar.com

- ▣  XFG Advanced Methods in Quantitative Finance
- ▣  Pricing Kernels and Risk Premia
- ▣  Hedging Cryptocurrency Options

### Code and data

- ▣ Seminar notebook (Colab); Quantlets: [github.com/danpele/MFM/tree/main/Quantlets/Ch\\_12](https://github.com/danpele/MFM/tree/main/Quantlets/Ch_12)

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